Tutorial 5 Solutions

1. The 95% CI for $\mu_1 - \mu_2$ is given by: $\bar{X}_1 - \bar{X}_2 \pm 1.96\sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}$

   where the subscripts denoted by a 1 represent gas and the subscripts denoted by a 2 represent electricity. This gives

   $160 - 180 \pm 1.96\sqrt{\frac{30^2}{50} + \frac{40^2}{60}}$

   Therefore we obtain the following 95% CI for $\mu_1 - \mu_2$: $(-33.10, -6.90)$.

   Since 0 is not included in the interval, it is likely that there is a difference between population means (more about this in hypothesis testing shortly).

2. (a) We can develop a 95% CI for σ^2 which is based on the observation that for *large* n, χ^2_{n-1} is **approximately** $\mathrm{N}(n-1, 2(n-1))$. Hence, we have

   $$\frac{\frac{(n-1)S^2}{\sigma^2} - (n-1)}{\sqrt{2(n-1)}} \doteq N(0,1).$$

   It follows that

   $$\mathbb{P}\left(-1.96 < \frac{\frac{(n-1)S^2}{\sigma^2} - (n-1)}{\sqrt{2(n-1)}} < 1.96\right) \doteq 0.95.$$

   Invert the inequalities to get

   $$\mathbb{P}\left(\frac{(n-1)S^2}{(n-1) + 1.96\sqrt{2(n-1)}} < \sigma^2 < \frac{(n-1)S^2}{(n-1) - 1.96\sqrt{2(n-1)}}\right) \doteq 0.95$$

   which yields the approximate CI:

   $$\left(\frac{(n-1)S^2}{(n-1) + 1.96\sqrt{2(n-1)}}, \frac{(n-1)S^2}{(n-1) - 1.96\sqrt{2(n-1)}}\right)$$

   (b) We can also develop a 95% confidence interval based on the approximation given in the statistical tables, i.e. for *large* n, $\sqrt{2\chi^2_{n-1}}$ is **approximately** $N(\sqrt{2n-3}, 1)$. Hence, we have

   $$\sqrt{2(n-1)\frac{S^2}{\sigma^2}} - \sqrt{2n-3}) \doteq N(0,1), \qquad \text{and}$$

   $$\mathbb{P}\left(-1.96 < \sqrt{2(n-1)\frac{S^2}{\sigma^2}} - \sqrt{2n-3} < 1.96\right) \doteq 0.95.$$

Invert the inequalities to get

$$\mathbb{P}\left(\frac{2(n-1)S^2}{(\sqrt{2n-3}+1.96)^2} < \sigma^2 < \frac{2(n-1)S^2}{(\sqrt{2n-3}-1.96)^2}\right) \doteq 0.95,$$

which yields the approximate CI:

$$\left(\frac{2(n-1)S^2}{(\sqrt{2n-3}+1.96)^2}, \frac{2(n-1)S^2}{(\sqrt{2n-3}-1.96)^2}\right).$$

(c) For $S^2 = 1$ and $n = 101$, an *exact* 95% CI (see class notes) is given by

$$\left(\frac{100}{\chi^2_{100,0.025}}, \frac{100}{\chi^2_{100,0.975}}\right) = (0.772, 1.347).$$

Using the result from part (a) above, we obtain an **approximate** 95% CI for σ^2 given by

$$\left(\frac{100}{127.7}, \frac{100}{72.28}\right) = (0.783, 1.384).$$

Finally, using the **approximate** 95% CI for σ^2 obtained in part (b) above, we have

$$\left(\frac{200}{258.14}, \frac{200}{147.54}\right) = (0.775, 1.356).$$

Note: The CI that we obtain by using the second approximation appears to give a better answer.

3. First compute the respective (observed) sample means and variances:

$$\begin{aligned} &\text{Control}: \quad \bar{x}_C = 13.4 \text{ and } s^2_C = 8.267 \\ &\text{Washed}: \quad \bar{x}_W = 12.1 \text{ and } s^2_W = 5.878. \end{aligned}$$

(a) Under the assumptions that the measurements are *normally distributed* and that the samples are *independent*, we know that

$$\frac{S^2_W/\sigma^2_W}{S^2_C/\sigma^2_C} = \frac{S^2_W \cdot \sigma^2_C}{S^2_C \cdot \sigma^2_W} \sim F_{9,9}$$

where S^2_C and S^2_W are the respective sample variances (before they are observed). It follows that

$$\mathbb{P}\big\{F_{9,9,0.975} < \frac{S^2_W \cdot \sigma^2_C}{S^2_C \cdot \sigma^2_W} < F_{9,9,0.025}\big\} = 0.95$$

and hence

$$\mathbb{P}\big\{\frac{S^2_C}{S^2_W}F_{9,9,0.975} < \frac{\sigma^2_C}{\sigma^2_W} < F_{9,9,0.025}\frac{S^2_C}{S^2_W}\big\} = 0.95.$$

Observe that $F_{9,9,0.025} = 4.033$ and $F_{9,9,0.975} = 1/F_{9,9,0.025}$, so for these data, a 95% (equal-tailed) CI for σ^C/σ^2_W is given by:

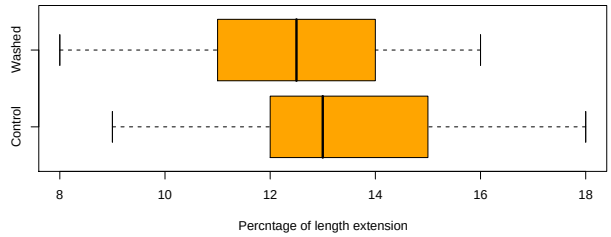
$$\left(\frac{8.267}{5.878}\frac{1}{4.033}, \frac{8.267}{5.878}4.033\right) = (0.35, 5.67).$$

(b) Under the assumptions that the measurements are *normally distributed*, the samples are *independent*, and the population variances σ_C^2 and σ_W^2 are *equal*, we know (see class notes) that the 95% (equal-tailed) CI for $\mu_C - \mu_W$ is given by

$$\left(\bar{X}_C - \bar{X}_W - t_{18,0.025} S_p \sqrt{\frac{1}{10} + \frac{1}{10}}, \bar{X}_C - \bar{X}_W + t_{18,0.025} S_p \sqrt{\frac{1}{10} + \frac{1}{10}}\right).$$

Since the sample sizes are *equal*, the pooled variance $S_p^2 = \frac{S_C^2 + S_W^2}{2}$, so for these data, the 95% CI for $\mu_C - \mu_W$ is given by $(-1.2, 3.8)$.

(c) The analysis in part (a) depends on the independence of the samples (which seems reasonable from the description of the experiment) and the normality of the population distributions. The normality assumption can be investigated by considering a plot of the respective samples:



Normality seems plausible. Equality of the variances (assumed in part (b)) also seems reasonable from the plot - but this assumption is also supported by the 95% CI calculated in part (a)!

Note that in practice it is often sensible to check such assumptions before performing any relevant analysis.

4. (a) First compute the respective (observed) sample means and variances:

Method A: $\sum x_i = 72.5, \ \sum x_i^2 = 595.81 \Rightarrow \bar{x}_A = 8.056$ and $s_A^2 = 1.4728$

Method B: $\sum x_i = 76, \ \sum x_i^2 = 650.7 \Rightarrow \bar{x}_B = 8.444$ and $s_B^2 = 1.1153.$

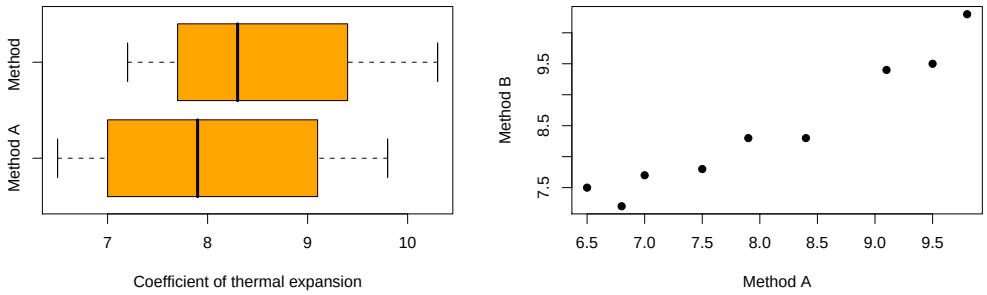
Since the sample sizes are equal, the pooled sample variance is given by

$$s_P^2 = \frac{s_A^2 + s_B^2}{2} = 1.2940.$$

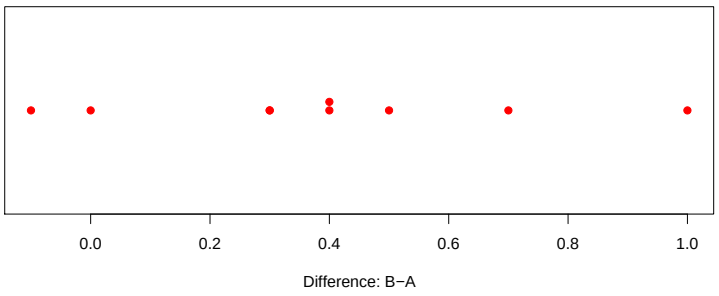
Assuming normality, equality of variances, and independence of the samples, the 95% CI for $\mu_B - \mu_A$ is given by

$$\left(\bar{x}_B - \bar{x}_A - t_{16,0.025} s_P \sqrt{\frac{2}{9}}, \ \bar{x}_B - \bar{x}_A + t_{16,0.025} s_P \sqrt{\frac{2}{9}}\right) = (-0.75, 1.525).$$

Next, check the underlying assumptions! Plots suggest that normality of the population distributions and equality of the respective variances are plausible, but a scatter plot of the Method B data against the Method A data suggests that it may be paired data:



(b) A plot of the differences $(d_1, d_2, ..., d_9) \equiv (x_{B,1} - x_{A,1}, ..., x_{B,9} - x_{A,9}) = (0.4, -0.1, 0.4, 1.0, 0, 0.7, 0.3, 0.3, 0.5)$ suggests that the normality assumption for the paired data is plausible.



For the differences we have $\bar{x}_D = 0.389$ and $s_D^2 = 0.111$, so a 95% CI for $\mu_D = \mu_B - \mu_A$ is given by

$$\left(0.389 - t_{8,0.025}\sqrt{\frac{0.111}{9}}, 0.389 + t_{8,0.025}\sqrt{\frac{0.111}{9}}\right) = (0.13, 0.65).$$

Note: since the data are paired, the first CI obtained above is not valid. Also, observe that the second interval is much narrower. This illustrates that paired data allow us to provide a more accurate estimate of $\mu_B - \mu_A$.

5. Let X denote the number of people in the sample who support the government's policy.

(a) Since the population size is large, we can assume that for a sample of size $n = 10$, the variable X has (approximately) a $\text{Bin}(10, p)$ distribution. Also, since the sample size is small, we can use the Binomial tables to determine a 95% CI (p_L, p_U) for p by solving the equations:

$$\mathbb{P}(X \geq 3 \mid p = p_L) = 1 - F(2; 10, p_L) = 0.025$$

$$\mathbb{P}(X \leq 3 \mid p = p_U) = F(3; 10, p_U) = 0.025.$$

To solve the first equation, we note that

$$\mathbb{P}(X \geq 3 \mid p = p_L) = 1 - F(2; 10, p_L) = 0.025 \Rightarrow F(2; 10, p_L) = 0.975$$

From the tables we obtain

$$F(2; 10, 0.06) = 0.9812 \text{ and } F(2; 10, 0.07) = 0.9717,$$

so (by interpolation) we obtain $p_L = 0.067$.

To solve the second equation, let Y denote a $\text{Bin}(10, 1-p)$ and observe that

$$0.025 = \mathbb{P}(X \leq 3 \mid p = p_U) = \mathbb{P}(Y \geq 7 \mid 1-p = 1-p_U) \Rightarrow F(6; 10, 1-p_U) = 0.975.$$

From the tables we obtain

$$F(6; 10, 0.34) = 0.9780 \text{ and } F(6; 10, 0.35) = 0.9740.$$

So (by interpolation) $1 - p_U = 0.3475 \approx 0.348$ and hence, $p_U = 0.652$ (to 3dp).

Conclusion: 95% CI for p is given by $(0.067, 0.652)$.

(b) For a sample of size $n = 100$, the variable X has (approximately) $\text{Bin}(100, p)$ distribution. In this case a (simple) 95% CI for p is given by

$$\left(\hat{p} - 1.96\sqrt{\frac{\hat{p}(1-\hat{p})}{n}}, \hat{p} + 1.96\sqrt{\frac{\hat{p}(1-\hat{p})}{n}}\right) = (0.21, 0.39)$$

(where $\hat{p} = \frac{30}{100} = 0.3$).

Comment: The CI obtained from a sample of size 100 is much narrower than the one obtained from a sample of size 10. This is due to the sample size - i.e. using more data yields a more accurate estimate.

Exam Style Question

Consider two normally distributed populations X and Y with means μ_X and μ_Y and variances σ_X^2 and σ_Y^2 respectively. Also consider two independent random samples, one from each population, given by $X_1, \ldots, X_{n_X}$ and $Y_1, \ldots, Y_{n_Y}$ respectively.

(a) Note that $\bar{X} - \bar{Y}$ is an unbiased estimator for $\mu_X - \mu_Y$ if

$$\mathbb{E}[\bar{X} - \bar{Y}] = \mu_X - \mu_Y.$$

By linearity of the expectation:

$$\mathbb{E}[\bar{X}] = \mathbb{E}\left[\frac{1}{n_X}\sum_{i=1}^{n_X} X_i\right] = \frac{1}{n_X}\sum_{i=1}^{n_X}\mathbb{E}\left[X_i\right] = \frac{1}{n_X}\sum_{i=1}^{n_X}\mu_X = \mu_X.$$

Similarly $\mathbb{E}[\bar{Y}] = \mu_Y$ then

$$\mathbb{E}[\bar{X} - \bar{Y}] = \mathbb{E}[\bar{X}] - \mathbb{E}[\bar{Y}] = \mu_X - \mu_Y.$$

Hence $\bar{X} - \bar{Y}$ is an unbiased estimator for $\mu_X - \mu_Y$.

[2 marks]

(b) For h to be a pivotal quantity we need:

- h is a function of both the sample and the parameter of interest;
- h has a known (or approximately known) distribution;
- h is monotonic.

The first condition follows sice h depends directly on the parameter of interest $\mu_X - \mu_Y$ and depends on the sample via $\bar{X} - \bar{Y}$.

To find the distribution of $h(\underline{X}, \underline{Y}, \mu_X - \mu_Y)$ we note that

$$\bar{X} \sim \mathrm{N}\left(\mu_X, \frac{\sigma_X^2}{n_X}\right), \quad \text{and} \quad \bar{Y} \sim \mathrm{N}\left(\mu_Y, \frac{\sigma_Y^2}{n_Y}\right).$$

Note that $\bar{X}$ and $\bar{Y}$ are independent and the sum of two normal random variables is normally distributed, so we have

$$\bar{X} - \bar{Y} \sim \mathrm{N}\left(\mu_X - \mu_Y, \frac{\sigma_X^2}{n_X} + \frac{\sigma_Y^2}{n_Y}\right).$$

Since the linear transformation of normal random variables is also normally distributed we have

$$\frac{\bar{X} - \bar{Y} - (\mu_X - \mu_Y)}{\sqrt{\frac{\sigma_X^2}{n_X} + \frac{\sigma_Y^2}{n_Y}}} \sim \mathrm{N}(0, 1).$$

Therefore the distribution of $h(\underline{X}, \underline{Y}, \mu_X - \mu_Y)$ is known and the second condition holds.

The third condition hold since $h(\underline{X}, \underline{Y}, \mu_X - \mu_Y)$ is a linear function of $\mu_X - \mu_Y$.

[4 marks]

(c) Since the variances are not known we will assume that the population variances are equal and use the pivotal quantity

$$h(\underline{X}, \underline{Y}, \mu_X - \mu_Y) = \frac{\bar{X} - \bar{Y} - (\mu_X - \mu_Y)}{S_p\sqrt{\frac{1}{n_X} + \frac{1}{n_Y}}}, \quad S_p^2 = \frac{(n_X - 1)S_X^2 + (n_Y - 1)S_Y^2}{n_X + n_Y - 2}.$$

Note that this has a known distribution of $t_{n_X+n_Y-2}$. Using the known distribution we have

$$\mathbb{P}\left(-t_{n_X+n_Y-2,0.025} \leq \frac{\bar{X} - \bar{Y} - (\mu_X - \mu_Y)}{S_p\sqrt{\frac{1}{n_X} + \frac{1}{n_Y}}} \leq t_{n_X+n_Y-2,0.025}\right) = 0.95.$$

Rearranging gives the equal-tailed 95% confidence interval as:

$$\left(\bar{X} - \bar{Y} - t_{n_X+n_Y-2,0.025}S_p\sqrt{\frac{1}{n_X} + \frac{1}{n_Y}}, \bar{X} - \bar{Y} + t_{n_X+n_Y-2,0.025}S_p\sqrt{\frac{1}{n_X} + \frac{1}{n_Y}}\right)$$

Using the data given we have:

$$\begin{aligned} n_X = 12, \quad n_Y = 10 \\ \bar{x} - \bar{y} = -5 \\ s_p^2 = \frac{(11)(4.5^2) + (9)(3.1^2)}{20} = 15.462 \end{aligned}$$

From the statistical tables we have $t_{20,0.025} = 2.086$. Therefore our observed confidence interval is

$$\left(-5 - 2.086\sqrt{15.462}\sqrt{\frac{1}{12} + \frac{1}{10}}, -5 + 2.086\sqrt{15.462}\sqrt{\frac{1}{12} + \frac{1}{10}} \right) = (-8.512, -1.488).$$

[4 marks]